Turning Churn Risk into Retained Revenue

A value-weighted retention proposal for Company A (Telecom)

Proof-of-Concept · Data-driven business proposal backed by a machine-learning model

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THE ONE-SLIDE TAKEAWAY

Executive summary

Company A can grow margin fastest by keeping the customers it already has.
I built a model that ranks every customer by churn risk and revenue at stake, so the retention team spends its budget only where it pays back.

0.70	79.5%	$2.07M	2.1×
Model ROC-AUC (XGBoost)	churn in the top-risk decile	net benefit / year	return on campaign spend

Recommendation: target the top 20% highest-risk, high-value customers with a handset-upgrade / bill-credit offer. This captures ~30% of all churners and is net-positive even under conservative assumptions.

MARKET CONTEXT

Why retention is the right lever

- Mobile markets like India are saturated — net new connections are scarce, so operators grow by lifting revenue-per-user and cutting churn, not by adding subscribers.
- Winning back a lost subscriber costs several times more than keeping an existing one — retention is the cheaper, higher-margin growth lever.
- Churn is not random: usage decline, ageing handsets and support friction show up months before a customer leaves — so churn is predictable, and therefore preventable.
- Implication for Company A: a model that flags at-risk, high-value customers early turns a reactive cost into a proactive, measurable revenue play.

Sources: Harvard Business Review (2014) [1]; TRAI Performance Indicators [2]; GSMA, The Mobile Economy [3]. See References.

STARTING POINT

The client and the data

- Company A — a wireless operator with ~100,000 customers and no in-house ML capability.
- Two linked tables joined 1:1 on Customer_ID into a single 100,000 × 100 analysis table.

Client.csv

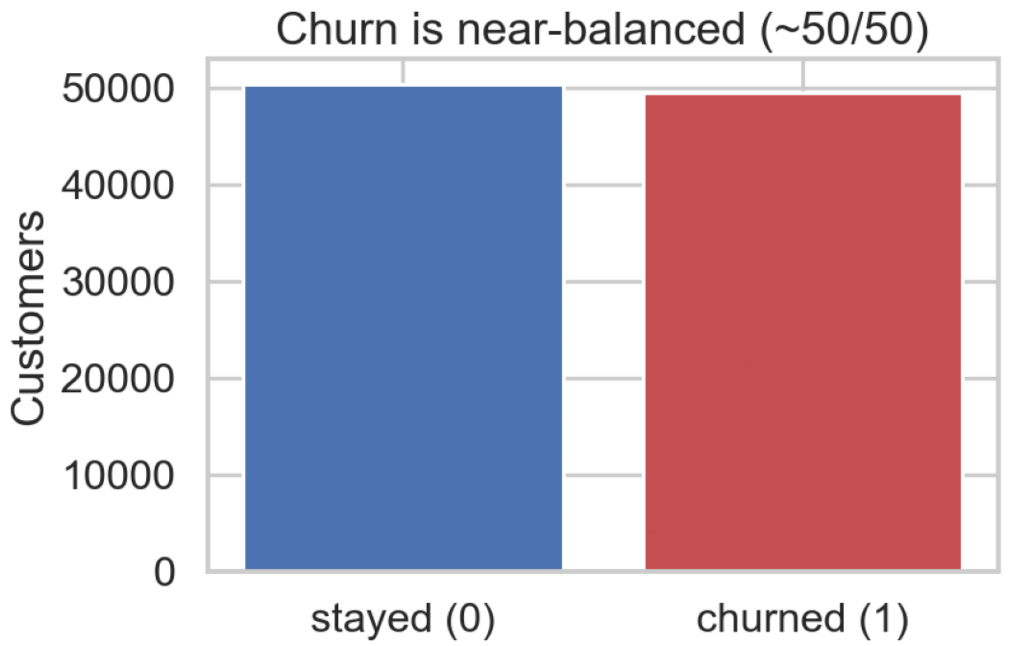
One row per customer:
• tenure, plan, credit class
• demographics
• equipment (handset age, price)

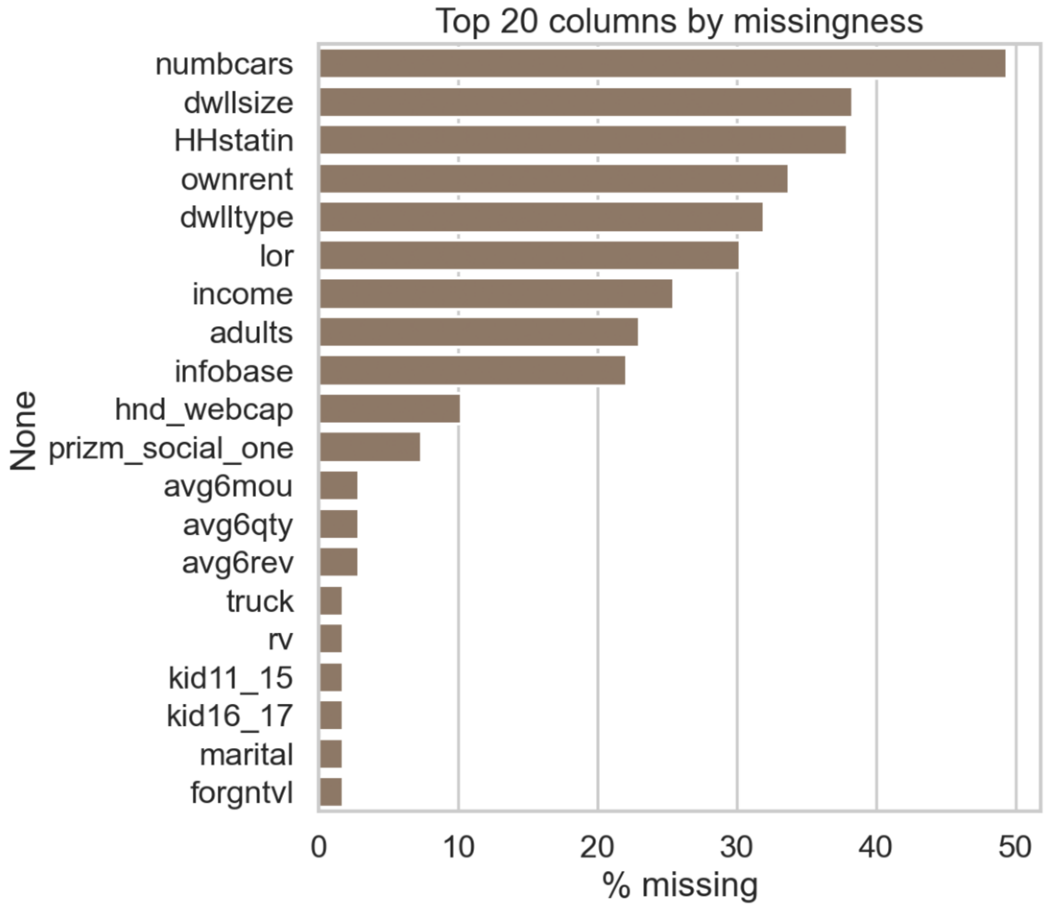
Record.csv

One row per customer:
• monthly usage & minutes
• billing & overage
• call quality + churn target

Target variable: churn = 1 if the customer left within 31–60 days of the observation date.

What the data is — and isn't — good for

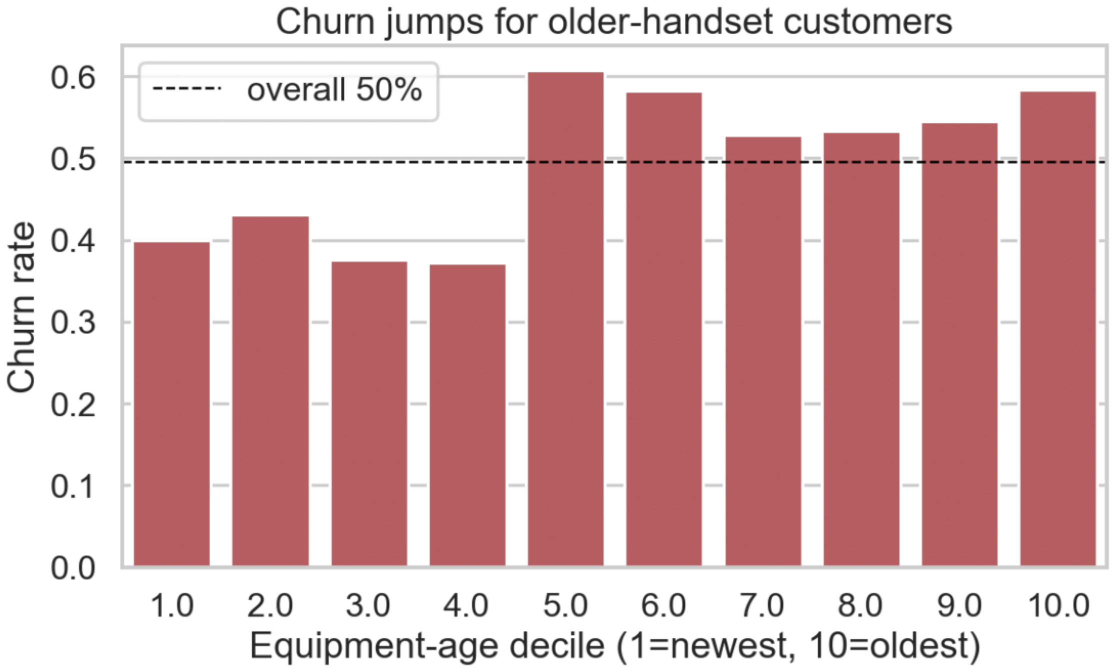


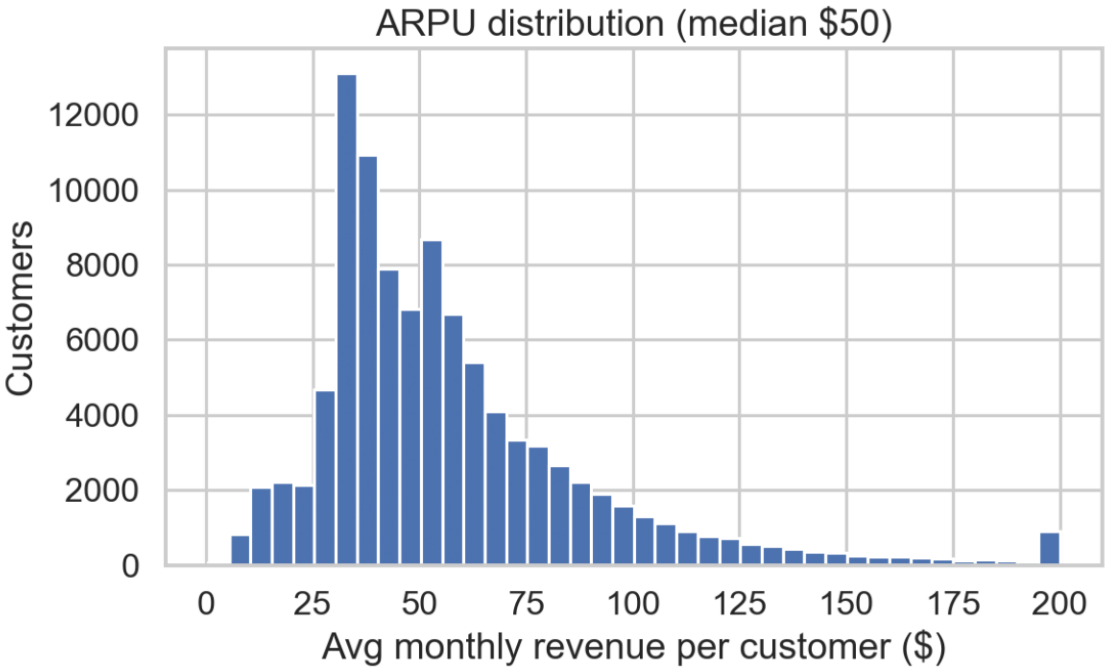


Churn is ~50/50 — accuracy is meaningful, but I rank by probability (ROC-AUC) for targeting.

Demographic columns are 30–50% missing (third-party append) and are dropped; operator-owned usage/billing data is near-complete and actionable.

The lever: ageing handsets drive churn





Newer-handset customers churn ~37–43%; older-handset customers ~53–60% — a clear, intervenable threshold.
Revenue is right-skewed, so a few high-ARPU customers carry outsized value.

Handset age is one of the few churn drivers the business can actually change (upgrade / subsidy) — unlike tenure or demographics.

Problem definition

Business question	Which customers should we spend retention budget on, and what is the return?
Target variable	churn (binary) — justified by EDA: balanced, and tied to an actionable lever.
ML task	Classification used as a ranking model — score every customer by churn probability.
Primary metric	ROC-AUC (threshold-free, suits ranking); Accuracy / F1 reported as support.
Who acts on it	Retention / CRM team — sends a targeted offer to the highest-risk, high-value list.

The model is evidence for a decision, not the deliverable itself — every choice above serves the retention recommendation.

METHOD

Approach: features and models

Feature engineering — encode business intuition no single column captures:

- handset age relative to tenure · price-per-minute · overage (bill-shock) share
- support-call intensity · recent usage-decline flag · share of active lines

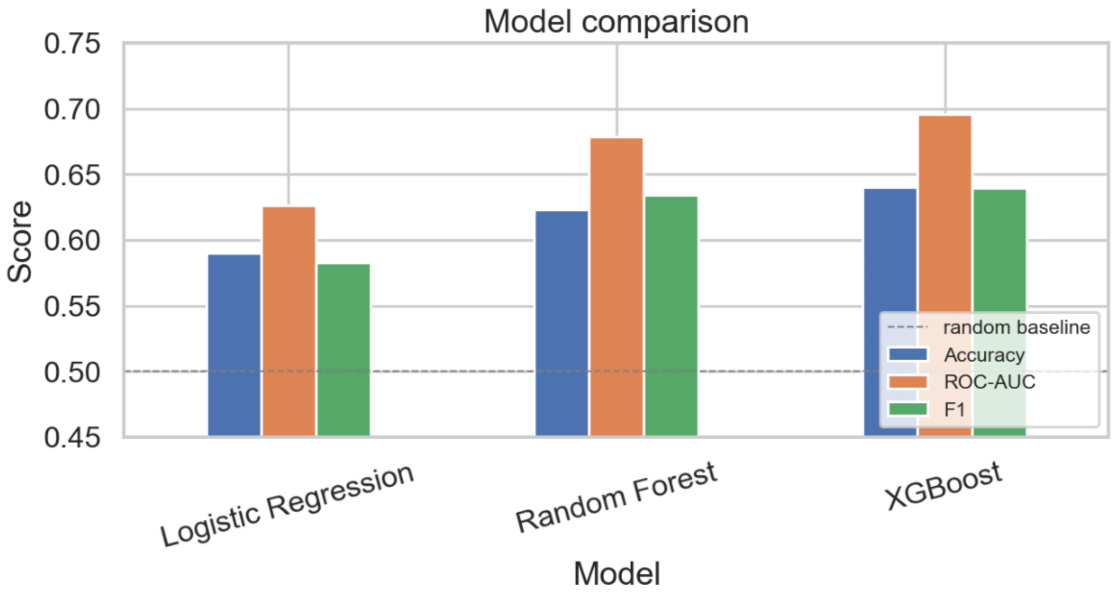
Models compared (weakest → strongest, so the choice is justified):

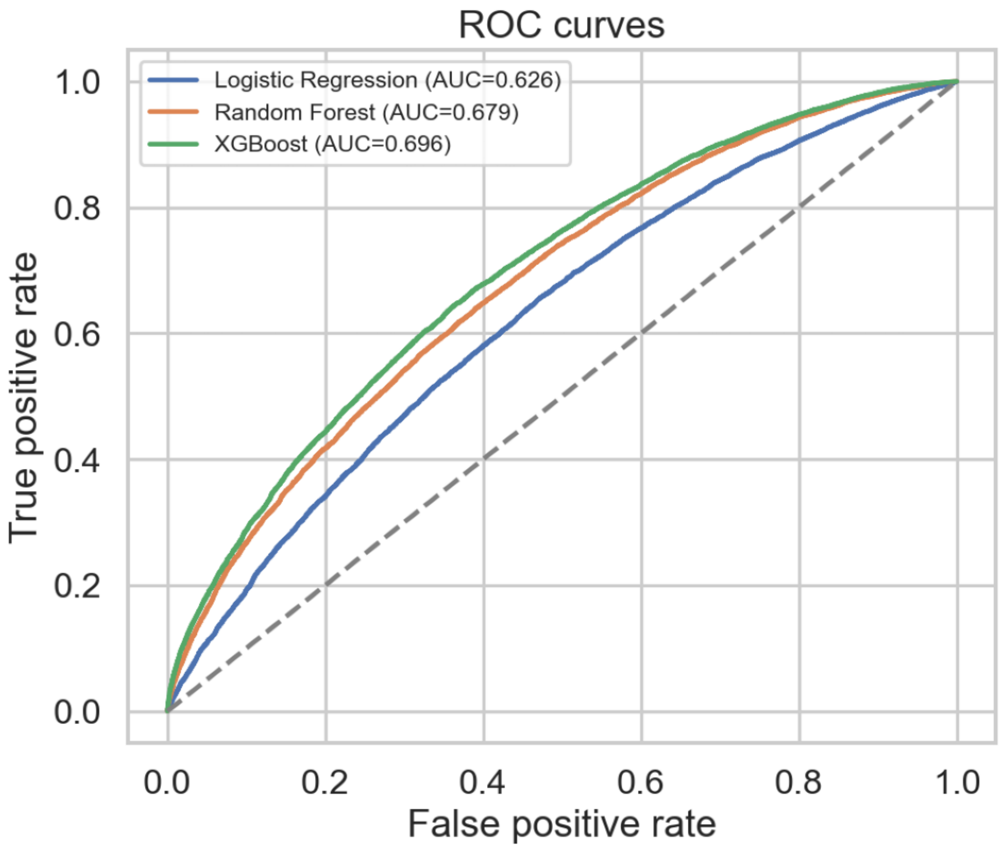
- **Logistic Regression** — interpretable linear baseline.
- **Random Forest** — captures non-linear interactions.
- **XGBoost** — gradient-boosted trees: strong on tabular data, handles missing values, explainable.

All models share the same stratified 70/30 train/test split and random_state = 42 for a fair, reproducible comparison.

EVALUATION

Model results — XGBoost wins





Held-out test scores

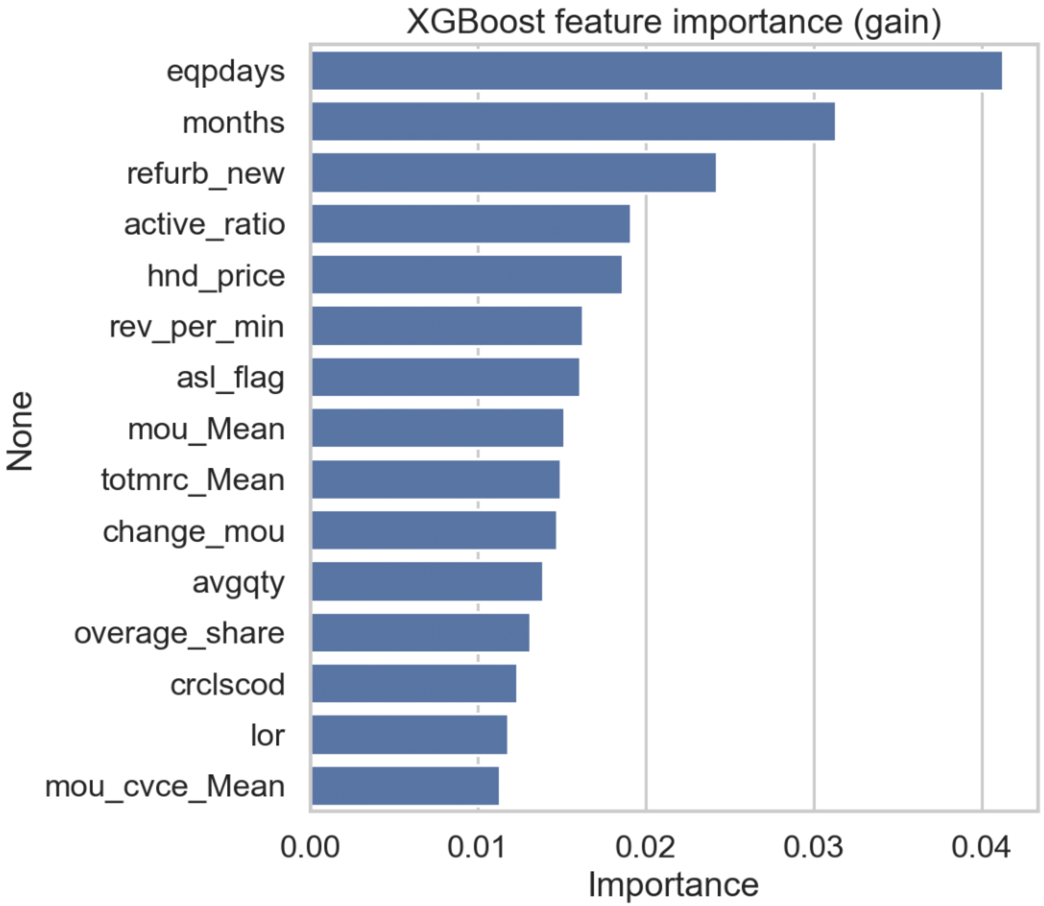
Model	Acc	AUC	F1
Logistic Reg.	0.59	0.626	0.58
Random Forest	0.62	0.679	0.63
XGBoost	**0.64**	**0.696**	**0.64**

Model: XGBoost · Metric: ROC-AUC = 0.696 (Accuracy 0.64, F1 0.64) on a 30% held-out test set.

Selected model: XGBoost — ROC-AUC 0.70 vs 0.50 random baseline.

INTERPRETATION

What the model says drives churn



- Equipment age (eqpdays, age-vs-tenure) is the strongest signal.
- Billing & usage level and price-per-minute follow.
- Support-call intensity flags service friction.

Importance + intervenability:

Tenure and demographics rank high but cannot be changed. Handset age can — so it becomes the lever the proposal pulls.

Feature importance = XGBoost 'gain' (how much each feature improves the model when used).

FROM SCORES TO A TARGET LIST

Targeting by risk works

Rank by risk, act on the top slice:

Risk decile	Churn rate	Lift
1 (highest)	**79.5%**	**1.60×**
2	67.8%	1.37×
3	61.4%	1.24×
10 (lowest)	19.3%	0.39×

The cumulative-gains curve shows the top 20% of customers by risk contain ~30% of all churners — far more than the 20% a random campaign would hit.

Lift = decile churn rate ÷ overall churn rate. A perfectly random target would have lift = 1.0 in every decile.

THE PROPOSAL IN DOLLARS

Quantified business impact

Campaign: contact the top 20% by risk on the full 100,000-customer book.

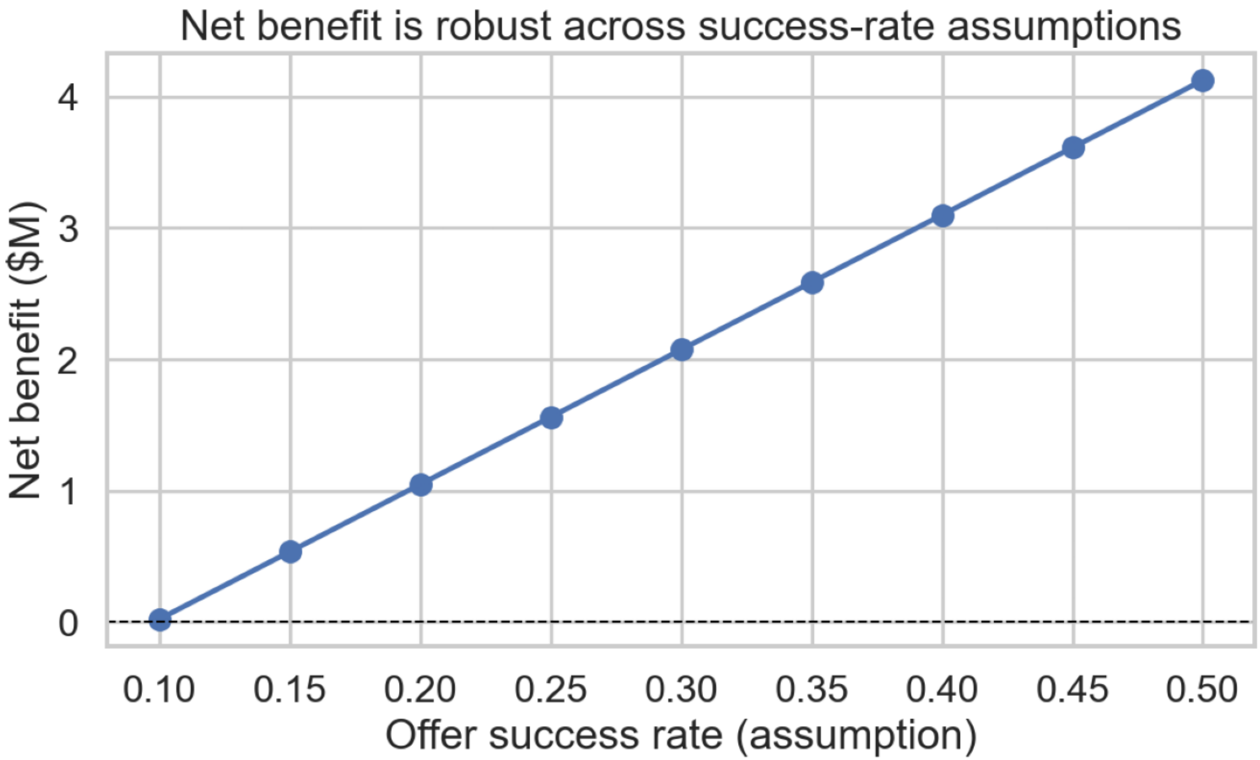
20,000	~4,400	$3.07M	$2.07M
customers contacted	churners actually saved	revenue retained	NET benefit

Assumptions (stated transparently, stress-tested next slide):

- Offer cost $50 / contacted customer · Offer success rate 30% · Value horizon 12 months
- Avg revenue of targeted churners ≈ $58 / month · Campaign cost $1.0M → 2.1× ROI

Net benefit = (saved customers × monthly ARPU × 12) − campaign cost. Targeting fraction is model-derived; cost & success rate are planning assumptions.

Is the case robust?



- The biggest unknown is the offer success rate.
- The campaign stays net-positive across the full plausible range (10–50%).
- Even at a pessimistic 10% success, the program does not lose money.
- So the recommendation is robust — not knife-edge on one assumption.

A real A/B pilot (Next Steps) would replace the assumed success rate with a measured one before full roll-out.

WHAT TO DO NEXT

Recommendation & roadmap

Recommendation

Score the customer book monthly; target the top-risk, high-ARPU segment with a handset-upgrade or bill-credit offer, prioritising customers also showing usage decline or support friction.

Roadmap

- **Pilot** — run a small A/B test to measure the true offer success rate.
- **Scale** — roll out monthly scoring; tune offer cost by ARPU tier.
- **Explain** — add SHAP per-customer reasons for the CRM team.
- **Sustain** — monitor for drift and re-train quarterly.

Risks: model shows correlation not causation; offer cannibalisation and seasonality are out of scope for the PoC.

References

[1] A. Gallo, “The Value of Keeping the Right Customers,” Harvard Business Review, 2014. https://hbr.org/2014/10/the-value-of-keeping-the-right-customers

[2] Telecom Regulatory Authority of India (TRAI), “Indian Telecom Services Performance Indicators.” https://www.trai.gov.in/release-publication/reports

[3] GSMA, “The Mobile Economy” (annual report series). https://www.gsma.com/mobileeconomy/

[4] T. Chen & C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” KDD ’16, 2016. https://doi.org/10.1145/2939672.2939785

[5] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” JMLR 12, 2011. https://jmlr.org/papers/v12/pedregosa11a.html

Dataset: Company A telecom dataset provided with the GCI World 2026 final assignment.